
Modern Algebra 1

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Problem 1

(a) Let $\alpha : A \rightarrow B$, $\beta : B \rightarrow C$, and $\gamma : C \rightarrow D$. It follows that

$$(\gamma(\beta\alpha))(a) = \gamma((\beta\alpha)(a)) = \gamma(\beta(\alpha(a))) = (\gamma\beta)(\alpha(a)) = ((\gamma\beta)(\alpha))(a).$$

Therefore, $\gamma(\beta\alpha) = (\gamma\beta)\alpha$.

(b) Let α, β be injective functions. Let $(\beta\alpha)(a_1) = (\beta\alpha)(a_2)$. We have

$$\beta(\alpha(a_1)) = \beta(\alpha(a_2)).$$

Since β is injective, it follows that $\alpha(a_1) = \alpha(a_2)$. Since α is injective, it follows that $a_1 = a_2$. Therefore, $\beta\alpha$ is injective.

(c) Let $\alpha : \mathbb{R} \rightarrow \mathbb{R}$ be the function $\alpha(x) = x$. Clearly α is injective. However, α is also clearly surjective, so we can define $\beta : \mathbb{R} \rightarrow \mathbb{R}$ as the map $\beta(x) = x^2$ and consider the function $(\beta\alpha)(x)$. It follows that

$$(\beta\alpha)(1) = \beta(\alpha(1)) = \beta(1) = 1^2 = 1.$$

Additionally, we have

$$(\beta\alpha)(-1) = \beta(\alpha(-1)) = \beta(-1) = (-1)^2 = 1.$$

Since $(\beta\alpha)(1) = (\beta\alpha)(-1)$ and $1 \neq -1$, $\beta\alpha$ is not injective.

(d) Let α, β be surjective functions, and let $c \in C$. Since $\beta : B \rightarrow C$ is surjective, there exists some $b \in B$ such that $\beta(b) = c$. Since α is surjective, there exists some $a \in A$ such that $\alpha(a) = b$. Therefore, we have

$$(\beta\alpha)(a) = \beta(\alpha(a)) = \beta(b) = c.$$

Therefore, $\beta\alpha$ is surjective.

(e) Let 0 be a specific element in B . Let $\beta : B \rightarrow C$ be any onto function, and let $\alpha : A \rightarrow B$ map $a \mapsto 0$ for all $a \in A$. It follows that for all $a \in A$,

$$(\beta\alpha)(a) = \beta(0).$$

For any C such that there is some $c \in C \setminus \{\beta(0)\}$ (in other words, there is more than one element in C), then $\beta\alpha$ is not onto because there exists some $c \in C$ such that $(\beta\alpha)(a) \neq c$ for all $a \in A$.

(f) Let $\alpha : A \rightarrow B$ be a bijection. Define $\alpha^{-1} : B \rightarrow A$ to be the map such that if $\alpha(a) = b$, then $\alpha^{-1}(b) = a$. Since α is surjective, all $b \in B$ have a corresponding $a \in A$ such that $\alpha(a) = b$. Since α is also injective, for any $b \in B$, there exists precisely one $a \in A$ such that $\alpha(a) = b$. Therefore, all $b \in B$ have precisely one corresponding $a \in A$ such that $\alpha(a) = b$, and thus α^{-1} is well defined. Therefore, for any given bijection α , such a map α^{-1} exists. It follows that

$$(\alpha^{-1}\alpha)(a) = \alpha^{-1}(\alpha(a)) = \alpha^{-1}(b) = a.$$

Additionally,

$$(\alpha\alpha^{-1})(b) = \alpha(\alpha^{-1}(b)) = \alpha(a) = b.$$

Problem 2

Let S have n elements, and let $\phi : S \rightarrow S$ be injective. Suppose there exists some $s^* \in S$ such that there does not exist $s \in S$ such that $\phi(s) = s^*$. This implies the image of ϕ has at most $n - 1$ elements. Thus, ϕ maps n elements to $n^* \leq n - 1$ elements. This implies there must exist at least two elements $s_1, s_2 \in S$, $s_1 \neq s_2$ such that $\phi(s_1) = \phi(s_2)$. This is a contradiction since ϕ is injective, implying that there does not exist such an $s^* \in S$ such that for all $s \in S$, $\phi(s) \neq s^*$. Therefore, ϕ is surjective.

Again let S have n elements and let $\phi : S \rightarrow S$ be surjective. Suppose there exists some $s_1, s_2 \in S$, $s_1 \neq s_2$ such that $\phi(s_1) = \phi(s_2)$. Since some element is mapped to by two elements of the domain, and since the domain and range are the same finite size, then there must exist at least one $s^* \in S$ such that for all $s \in S$, $\phi(s) \neq s^*$. This contradicts the fact that ϕ is surjective, implying that for all $s_1, s_2 \in S$ with $s_1 \neq s_2$, then $\phi(s_1) \neq \phi(s_2)$. Therefore, ϕ is injective.

These statements do not hold for an infinite S . For example, take $\phi : \mathbb{Z} \rightarrow \mathbb{Z}$ defined as $\phi(n) = 2n$. Obviously, the image of ϕ is the set of even integers, and since this is a proper subset of $\mathbb{Z}$, ϕ is not surjective. However, ϕ is very much injective. Suppose $\phi(n) = \phi(m)$. It follows that

$$2n = 2m \implies n = m.$$

Therefore, ϕ is an example of an injective, but not surjective, map on an infinite set.

Problem 3

Let $\sigma = (a_1, \dots, a_m)$. Show $\sigma^i(a_k) = a_{k+i \pmod m}$ and $|\sigma| = m$.

We will show the first property via induction. Since $k + i$ is taken modulo m , it suffices to show the properties hold for all $i \in \mathbb{Z}_m$, since for any $i \in \mathbb{Z}$, there exists some $j \in \mathbb{Z}_m$ such that $i \equiv j \pmod m$. We know that $\sigma(a_k) = a_{k+1}$. First, let $i = 1$. We have

$$\sigma^1(a_k) = \sigma(a_k) = a_{k+1} = a_{k+i}$$

by definition. Now, let $\sigma^i(a_k) = a_{k+i}$ for an arbitrary $i \in \mathbb{Z}_m$. It follows that

$$\sigma^{i+1}(a_k) = \sigma(\sigma^i(a_k)) = \sigma(a_{k+i}) = a_{k+i+1}.$$

By the method of induction, $\sigma^i(a_k) = a_{k+i}$ for all $i \in \mathbb{Z}_m$, where $k + i$ is taken modulo m since $a_m \mapsto a_1$, so $\sigma(a_m) = \sigma(a_1) = a_2$, and so on.

Now we show $|\sigma| = m$. Let $\sigma^k(a_i) = a_i$. Then $i + k \equiv i \pmod m$. The minimum positive integer k such that $i + k \equiv i \pmod m$ is $k = m$. Therefore, $|\sigma| = m$.

Problem 4

Let $\alpha \in S_n$ be a cycle of length n . Prove $C(\alpha) = \langle \alpha \rangle$.

First, we show every $\alpha^k \in C(\alpha)$. We have

$$\alpha \alpha^k = \alpha \left(\alpha^{k-1} \alpha \right) = \left(\alpha \alpha^{k-1} \right) \alpha = \alpha^k \alpha.$$

Therefore, $\langle \alpha \rangle \subseteq C(\alpha)$.

Now we show the converse. Let $\sigma \in S_n$ such that $\sigma \alpha = \alpha \sigma$. Without loss of generality, define $\alpha = (12 \dots n)$. We then have the convenient representation $\alpha(i) = i + 1$. We have

$$(\sigma \alpha)(i) = \sigma(\alpha(i)) = \sigma(i + 1).$$

Since $\sigma \alpha = \alpha \sigma$, we have

$$(\alpha \sigma)(i) = \alpha(\sigma(i)) = \sigma(i) + 1.$$

Since $\sigma(i) + 1 = \sigma(i + 1)$ **for all** i , σ must shift every component by the same amount, and thus $\sigma(i) = i + d$ for some $d \in \{1 \dots n\}$. Therefore, $\sigma = \alpha^d$, and thus $\sigma \in \langle \alpha \rangle$. Therefore, $C(\alpha) \subseteq \langle \alpha \rangle$.

Therefore, $C(\alpha) = \langle \alpha \rangle$.

Problem 5

We can use proof by example. Consider $\sigma = (12)(13) = (132)$. Then $\sigma^2 = (123)$. Additionally, consider $\tau = (12)(14) = (142)$. Then $\tau^2 = (124)$. We have $\sigma^2\tau^2 = (123)(124) = (13)(24)$. Thus, $\sigma^2\tau^2$ is a product of disjoint transpositions. Since $\sigma^2\tau^2$ is a cycle on 4 distinct elements, it must be the square of some π which is also a cycle on 4 distinct elements. Therefore, π must be either the product of two disjoint transpositions, or a 4-cycle. However, consider the 4-cycle $(abcd)$. Notice that

$$(abcd) = (ad)(ac)(ab).$$

Since a 4-cycle can be written as a product of an odd number of transpositions, no 4-cycle can be in A_4 . Therefore, π must be the product of two disjoint transpositions. However, since a transposition is its own inverse and since disjoint cycles commute, for any product of disjoint transpositions $(ab)(cd)$ we have

$$((ab)(cd))^2 = (ab)(cd)(ab)(cd) = (ab)(ab)(cd)(cd) = e.$$

Therefore, there exists no such $\pi \in A_4$ such that $\pi^2 = (13)(24)$. Therefore, H is not closed for at least the two elements shown here, and is thus not a group.